

GaN Module Design Recommendations Based on the Analysis of a Commercial 3-Phase GaN Module

John Alex Brothers
Mainstream Engineering Corporation
200 Yellow Place
Rockledge, Florida, Brevard, 32955
jbrothers@mainstream-engr.com

Troy Beechner
Mainstream Engineering Corporation
200 Yellow Place
Rockledge, Florida, Brevard, 32955
tbeechner@mainstream-engr.com

Abstract—In this paper, a detailed analysis of a commercial 3-phase, 100 V, 270 A (per phase) GaN module is presented. A high frequency parasitic circuit model was derived for the module, which was simulated to demonstrate a unique sensitivity of low voltage GaN modules: its susceptibility to V_{DS} overshoot at hard turn-off switching transients due to large currents, fast GaN switching, large package parasitics, and low V_{DS} overshoot margins of 100 V rated dies. Hardware double pulse test (DPT) results confirmed this when the commercial module experienced premature failure due to V_{DS} overshoot when hard switched at 36% of its current rating. Module design layout recommendations are provided based on an analysis of the commercial GaN module's layout (using 3D electromagnetic software) as well as GaN module layouts from existing literature. The paper concludes with hardware DPT results of a custom 100 V, 360 A GaN module. The hardware was successfully hard switched at 375 A, demonstrating the feasibility of low voltage (≤ 100 V), high current (> 300 A) GaN modules capable of hard switching at their die current ratings.

Keywords—Gallium Nitride (GaN), GaN module, parasitics, module layout, paralleling, Double Pulse Test (DPT)

I. INTRODUCTION

For low voltage (< 100 V), high current (> 300 A) power converters, GaN based power modules seem to be a promising technology. Compared to Si, GaN can achieve significantly increased switching frequency operation due to its lower switching losses, near-zero reverse recovery, and smaller gate charge. With proper design, GaN has the potential to achieve higher efficiency and power density for these applications. However, as the power electronics community transitions from Si to GaN, care must be taken during the design of the power modules, as design techniques that may have been acceptable for Si based power modules can detriment the operation of a GaN based module. Most notably, large package parasitic inductances can cause premature failure of the GaN dies due to V_{DS} overshoot at hard switching turn-off transients. This is particularly true for the low voltage GaN dies due to high currents, fast GaN switching transients, large package parasitics, and low V_{DS} overshoot margins of ≤ 100 V rated dies. Other negative impacts that parasitics can introduce include increased losses (circulating resonant currents, large gate/power inductive loops require slower switching), increased EMI noise, reduced short circuit survivability, and increased risk of gate false turn-on.

Thus, the goal of this paper is to 1) demonstrate the practical limitations and detriments caused by package parasitics for GaN modules, 2) recommend design strategies to minimize package

parasitics, and 3) demonstrate the feasibility of low voltage high current GaN modules. This is achieved by analyzing a commercial GaN module (100 V, 3-phase, 270 A per phase), existing modules from literature, and a custom GaN design (100 V, 2-phase, 180 A per phase).

In Section II, a commercial GaN module is analyzed. A parasitics model was developed based on experimental impedance measurements. This model was simulated using LTSpice, suggesting premature failure of the dies due to V_{DS} overshoot caused by the package parasitics. This prediction was verified by configuring the module in a double pulse test (DPT) and switching it at increasing currents until failure. Test results confirmed premature failure when hard switching at 36% of its current rating (failed at 292 A, rated for 810 A based on die current rating). Options to work with the existing module are explored, revealing that the route for optimal performance would be a redesign of the power stage with improved parasitics.

In Section III, the module's layout is analyzed using 3D electromagnetic modeling software Computer Simulation Technology (CST). Design recommendations are given based on an analysis and critique of the commercial GaN module as well as an analysis of GaN modules from existing literature.

In Section IV, a custom GaN module (100 V, 2-phase, 180 A per phase) is implemented in a DPT to demonstrate the feasibility of low voltage high current GaN modules. The hardware was successfully hard switched up to 375 A. Compared to the commercial module, this module demonstrated a 35% increase in hard switching current capability (375 A compared to 277 A) with less V_{DS} overshoot (109.5 V compared to > 137.5 V). Additionally, this module used smaller gate resistors (4 Ω per die compared to 6.8 Ω per die) and had less than half the number of parallel dies (4 parallel GS61008T dies compared to 9).

II. COMMERCIAL GAN MODULE CHARACTERIZATION

A. Parasitics Model Extraction

In this section, the extraction of a detailed parasitics model for a commercial 3-phase GaN power module is presented. Fig. 1 shows the module and its equivalent circuit, not including parasitics. There are three half bridge phases, each containing six GS61008T dies, six 6.8 Ω gate resistors, and one 0.1 μ F decoupling capacitor. Given the 90 A rated GS61008T dies, each phase's theoretical current rating is 270 A. The five large tabs are the DC inputs and three switching nodes, and the smaller pins are connections for the six device gates, source returns, and internal thermocouple.

This work was supported by the US Army CCDC Ground Vehicle Systems Center under SBIR contract number W56HZV-18-C-0021.

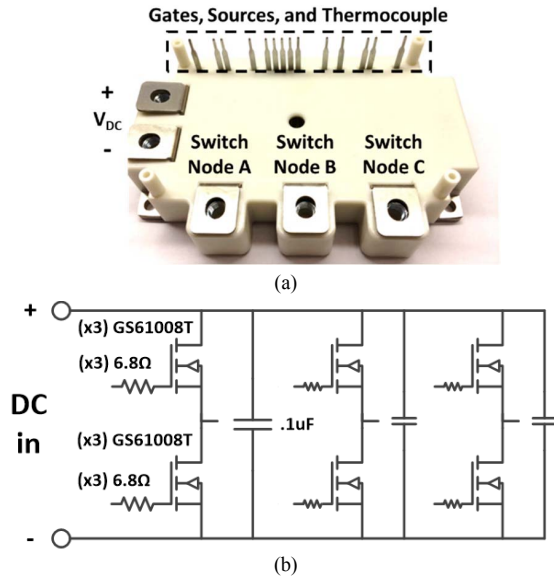


Fig. 1. (a) Commercial GaN module (3-phase, 100 V, 270 A per phase)
(b) schematic of the commercial GaN module

Fig. 2 shows the proposed high frequency differential mode equivalent circuit model. For simplicity, parallel dies within each phase are lumped into a single switch. Thus, C_{DS} , C_{GD} , and C_{GS} account for three dies in parallel and R_g is equivalently three $6.8\ \Omega$ resistors in parallel. Unlabeled components are the same across high and low side devices as well as across the three phases.

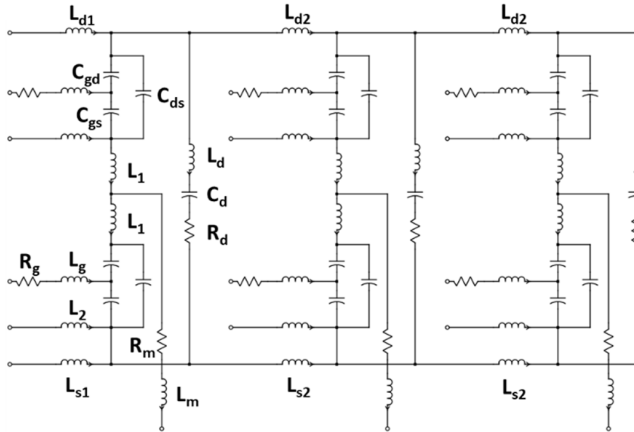


Fig. 2. Developed model of GaN module including packaging parasitics

TABLE I. SUMMARY OF MODULE PARASITIC ELEMENTS

Component	L _d	L _m	L _g	L ₁	L ₂	L _{d1}	L _{d2}	L _{s1}	L _{s2}
Unit	nH								
Value	5	44.6	18.8	0.35	6	13.8	12.8	10.3	9.9

Component	C _d	C _{DS}	C _{GS}	C _{GD}	R _d	R _m	R _g
Unit	nF				Ω		
Value	95.7	1.86	1.52	0.49	0.019	0.035	2.34

To derive the equivalent parasitics model, techniques from [1] through [3] were used. Table II outlines the first four impedance measurements taken on the module using an Agilent E4990A RF impedance analyzer, providing a capacitance, inductance, and resistance for each measurement. Based on the equivalent circuit model, equations were generated to relate components in the equivalent model to the measured values. For example, (1) shows the measurement 1 equivalent capacitance equation. With four equations (from equivalent circuit model), four solutions (from four measurements), and four unknowns (C_{DS} , C_{GS} , C_{GD} , and C_d), Mathematica was used to solve for C_{DS} , C_{GS} , C_{GD} , and C_d . The results are provided in Table I. Note: in (1), C_{GAN} is a single lumped capacitance accounting for C_{DS} , C_{GS} , and C_{GD} .

$$C_{meas1} = \frac{C_{GS}C_{GAN} + \frac{C_{GD}C_{DS}}{C_{GD} + C_{DS}}C_{GAN}}{C_{GS} + \frac{C_{GD}C_{DS}}{C_{GD} + C_{DS}} + C_{GAN}} + 3C_d + C_{GAN} \quad (1)$$

TABLE II. SUMMARY OF IMPEDANCE MEASUREMENTS 1 THROUGH 4

Measurement #	Node 1	Node 2
1 (All devices)	Gate	Source
2 (All phases)	High-side source	Low-side gate
3 (All phases)	High-side source	Low-side source
4 (All phases)	High-side gate	Low-side gate

TABLE III. EQUIVALENT EQUATIONS OF MEASUREMENTS 1 THROUGH 4

Equivalent Capacitance	Equivalent Inductance	Equivalent Resistance
Equation (1)	$L_g + L_2$	$R_g + R_2$
$F(C_{DS}, C_{GS}, C_{GD}, C_d)$	$L_2 + L_g + \frac{2L_1L_d}{2L_1 + L_d}$	$R_g + 2R_1 + R_2$
$F(C_{DS}, C_{GS}, C_{GD}, C_d)$	$2L_2 + \frac{2L_1L_d}{2L_1 + L_d}$	$2R_1 + 2R_2$
$F(C_{DS}, C_{GS}, C_{GD}, C_d)$	$\frac{2L_1L_d}{2L_1 + L_d} + 2L_g$	$2R_g + 2R_1$

Measurements 1 through 4 were also used to derive parasitic elements L_g , L_d , L_1 , L_2 , R_g , R_d , R_1 , and R_2 . For measurements 2 through 4, only a single L_d was considered in parallel with L_1 because the parallel inductive paths through L_{d2} and L_{s2} were so large compared to L_1 that they were considered negligible. R_1 and R_2 are in series with L_1 and L_2 , but were not included in the final parasitics model due to their relative insignificance. The final parameter values are provided in Table I.

In a similar fashion, measurements 5 through 7 were used to derive L_{d1} , L_{d2} , L_{s1} , L_{s2} , L_m , and R_m , whose results are included in Table I. A final 8th measurement of the module's input impedance was taken to observe the module's resonances due to its trace inductances, internal decoupling capacitors, and die capacitances, as well as verify the theoretical model. Measurements 5 through 8 are described in Table IV.

TABLE IV. SUMMARY OF IMPEDANCE MEASUREMENTS 5 THROUGH 8

Measurement #	Node 1	Node 2
5 (All phases)	High-side source	Switch node
6 (All phases)	Low-side gate	Negative DC input
7 (All phases)	High-side gate	Positive DC input
8	Positive DC input	Negative DC input

Fig. 3 shows the developed theoretical model's input impedance superimposed on the measured input impedance of the studied module, demonstrating strong agreement between the theoretical model and the hardware up to about 40 MHz. Comparable agreement was found for all measurements. Note the minimum impedance at 1.4 MHz caused by resonance between the internal decoupling capacitors C_d and trace inductances L_d , L_{d1} , L_{d2} , L_{s1} , and L_{s2} . This low frequency underdamped resonance is unique compared to conventional modules because it is caused by the use of local internal decoupling capacitors inside the module. Consequently, this resonance is easily excited during switching transients, which will be further explored in following sections.

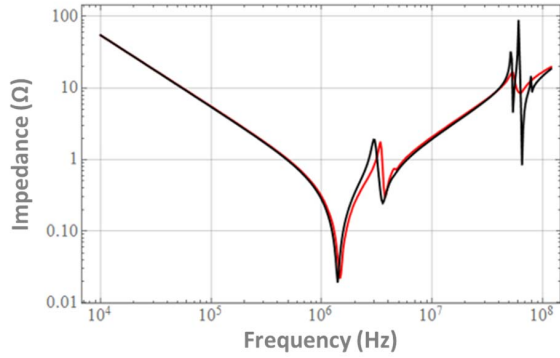


Fig. 3. Input impedance of GaN module (red) experimental measurement (black) equivalent circuit model

B. Performance Impacts Due to Package Parasitics

In this section, the various performance impacts and limitations caused by the module's parasitics are explored by simulating the derived model from Fig. 2. The module's model was simulated using LTSpice in a DPT configuration with all three switch nodes tied together, thus operating the module as a single half bridge with nine parallel GS61008T dies. Physics based device models were provided by the device manufacturer. Key simulation parameters include: $C_{bulk} = 500 \mu F$, $L = 4 \mu H$, $V_{DC} = 48 V$, $R_{g(on)} = 54.8 \Omega$ (per die), $R_{g(off)} = 6.8 \Omega$ (per die), $V_{GS(on)} = 6.2 V$, and $V_{GS(off)} = -2.8 V$.

Fig. 4 shows the simulated DPT waveforms. As seen, the third phase's low side dies experience a V_{DS} overshoot reaching 140 V when hard switched at 110 A. Based on the GS61008T datasheet [4], the dies can withstand an absolute maximum V_{DS} transient pulse of 130 V for up to 1 μs . Thus, the simulation was not pushed to higher currents.

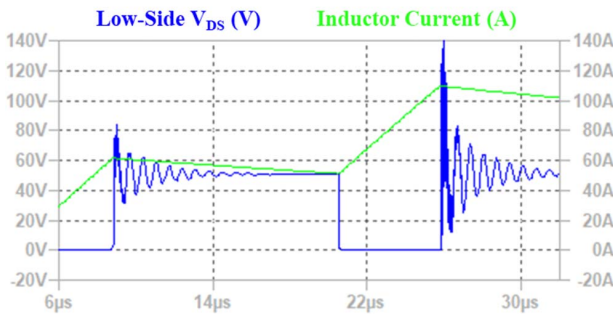


Fig. 4. DPT waveforms of the simulated parasitics model

Considering that the module is operating as a half bridge with nine parallel GS61008T dies (each die rated for 100 V, 90 A), the module's theoretical current rating is 810 A. Since simulation suggests that the dies exceed their V_{DS} rating when hard switching at 110 A, V_{DS} overshoot is expected to be the major performance limiting factor. Although more work is needed to determine the actual die failure point, it is predicted that the dies will fail well below the 810 A module rating. This also limits the survivable short circuit current unless additional soft turn-off protection features are implemented.

There are several other performance impacts that the module's parasitics introduce. Fig. 5 shows a zoomed in view of the simulation's second DPT pulse. As seen on the V_{DS} signal, there is a large underdamped 1.4 MHz oscillation. This is caused by resonance between the internal decoupling capacitors C_d and trace inductances L_d , L_{d1} , L_{d2} , L_{s1} , and L_{s2} . During operation, this resonance is excited at switching transients and induces circulating currents (causing unwanted conduction losses) and voltage ripple on the input and output buses. This voltage ripple is a major contributor to the die V_{DS} overshoot (the primary current limiting factor for the module) and causes increased EMI noise which could require additional filtering to meet commercial or military EMI standards.

A second oscillation can also be seen in Fig. 5 at approximately 80 MHz. This is primarily caused by resonance between the die off-state capacitances C_{DS} , C_{GS} , and C_{GD} and parasitic inductances L_l , L_d , however, L_{d1} , L_{d2} , L_{s1} , and L_{d2} also have an influence. This resonance is the second major contributor to V_{DS} overshoot (limiting the module's current capability) as well as causes increased switching losses.

Another impact of the package parasitics is the inability to accurately measure the die V_{DS} . Notice how in Fig. 5 the die V_{DS} reaches 140 V yet the voltage at the module terminals peaks at only 85 V. This is because the package parasitics act as impedance between the dies and the module terminals. Thus, whatever V_{DS} is measured during hardware testing, the true die V_{DS} is larger.

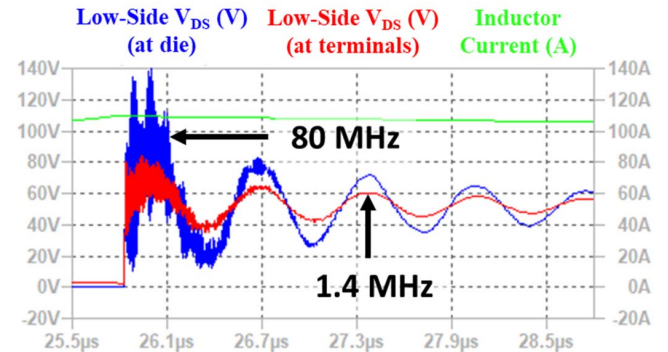


Fig. 5. Zoomed in view of the DPT simulation's second pulse

Another potential negative impact of this design relates to the gate. The large gate loop inductance (L_g and L_2 combine to 24.8 nH) and large 6.8 Ω internal gate resistor (per die) can reduce the pull-down capability of the gate driver which increases the chances of false turn-on. This can be combatted by increasing the turn-on gate resistance, however, this increases turn-on switching losses [5].

C. Hardware Testing of Commercial GaN Module

To validate the claims of reduced switching current capability, the hardware DPT setup shown in Fig. 6 was assembled. Four-layer PCBs were developed for the gate drive circuitry and external decoupling capacitors, using interleaving flux canceling layers to provide low inductance gate and power loop connections. The decoupling capacitor board had 154 μF of ceramic capacitors and 400 μF of film capacitors. Key hardware parameters were: $L = 4 \mu\text{H}$, $V_{\text{DC}} = 48 \text{ V}$, $R_{\text{g(on)}} = 54.8 \Omega$ (per die), $R_{\text{g(off)}} = 6.8 \Omega$ (per die), $V_{\text{GS(on)}} = 6.2 \text{ V}$, and $V_{\text{GS(off)}} = -2.8 \text{ V}$.

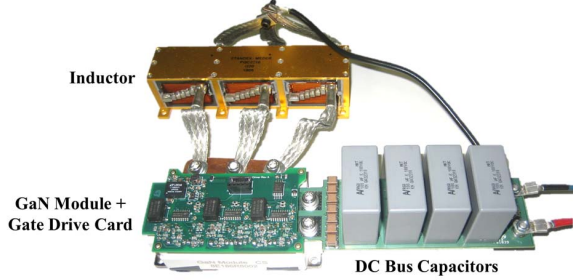


Fig. 6. Hardware DPT setup using the commercial GaN module

The module was subjected to increasing DPT currents until it experienced catastrophic failure. Fig. 7 shows the module hard switching at 277 A which was the test point before failure. The low side die V_{DS} was measured at the module terminals (actual die V_{DS} is larger). This V_{DS} measurement reached a peak of 130 V and had 2.1 MHz and 70 MHz components. The higher frequency 70 MHz oscillation matches the simulated 80 MHz reasonably well. The hardware's resonant frequency could be lower due to added inductance from the external capacitor PCB and/or the model's limited accuracy of the module's internal distributed inductances. The lower frequency resonance at 2.1 MHz is higher than the simulated 1.4 MHz because the hardware's internal capacitance decreased with the 48 V DC bias, a common trait of ceramic capacitor dielectrics.

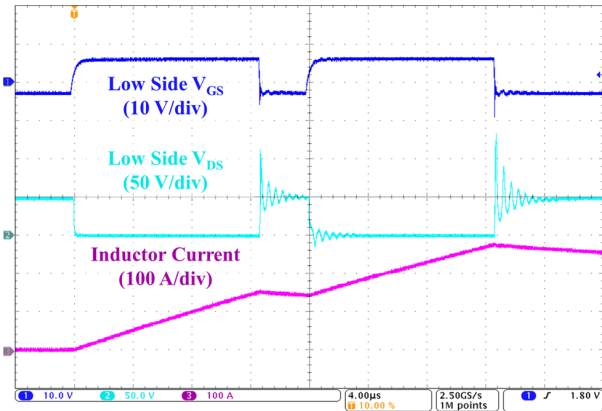


Fig. 7. Hardware DPT waveforms of the GaN module at 277 A

Finally, Fig. 8 shows the waveforms that resulted in module failure. When hard switched at 292 A, the measured V_{DS} overshoot reached a peak voltage of 137 V (at the module terminals) which irreversibly damaged the internal dies. This is why the 2.1 MHz V_{DS} oscillation is not present after the second pulse of this DPT.

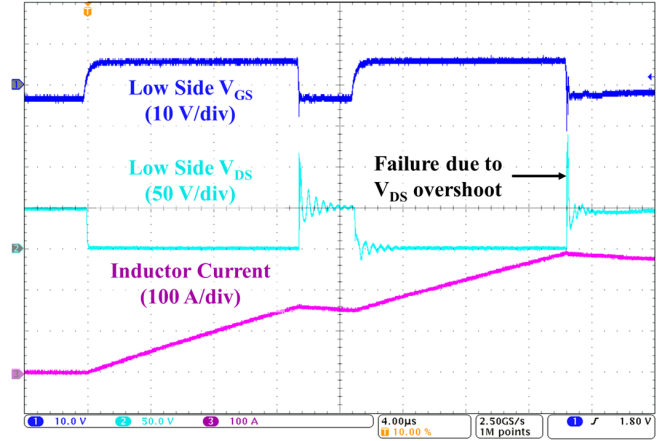


Fig. 8. Hardware DPT waveforms at 292 A resulting in module failure

D. The Necessity for Improved Parasitics

In this section, methods to improve the switching current capability of the existing module are presented, as well as their corresponding tradeoffs. This is used to emphasize why the only way to achieve optimal all-around performance is by improving the module's parasitics.

To improve the switching current capability of this module, the factors that dictate V_{DS} overshoot must first be understood. In this scenario, four factors contribute to the V_{DS} overshoot: 1) switching current amplitude (per die), 2) DC bus voltage, 3) switching speed, and 4) circuit parasitics.

The switching current amplitude and DC bus voltage are typically dictated by the application and topology, which makes them difficult to change. One way to reduce these is with the use of a unique topology which splits the current between more dies (such as using a full bridge instead of a half bridge), or subjects the dies to less than the DC bus voltage (such as the topology presented in [6]). These can come at the expense of increased die count and/or increased layout and control complexities. Another method to decrease the die switching current is to parallel multiple modules at the expense of increased size, cost, and complexity.

A simpler way to decrease the V_{DS} overshoot at hard turn-off is to decrease the switching speed by increasing the gate turn-off resistor value. This decreases the di/dt in the parasitic inductances and thus decreases the V_{DS} overshoot. This comes at the expense of increased turn-off switching losses [5] as well as diminished gate driver decoupling/pull down capability. This increases the risk of false turn-on and shoot-through when the complementary device(s) turn-on. This false turn-on can be mitigated by increasing the turn-on resistors (at the expense of increased turn-on switching losses), adding parallel capacitance to C_{GS} (decreases switching speed which gives increased switching losses, increased gate drive losses), implementing miller clamping (increased gate drive complexity), and/or by using a negative off gate bias (increased reverse conduction losses, increased gate drive complexity).

The final factor that affects the circuit's V_{DS} overshoot is the circuit parasitics, which are unchangeable for a preassembled module. However, one way to effectively alter the resonant impedance of the module is to add a snubber network. A snubber such as an RC damper could be added to the module's DC input and/or between the switching nodes and one of the DC input terminals. This method could be useful for damping the 2.1 MHz resonant component found in the hardware testing but practically cannot damp the 70 MHz component due to parasitic inductances between the 70 MHz resonant loop and the damper. Additionally, this approach adds unnecessary losses, size, and complexity.

Another possibility for altering the module's resonant structure is to increase the internal capacitance value (not a possibility from the user side, only from the manufacturer side). This would effectively lower the resonant frequency between the external and internal capacitors and increase the damping factor of this RLC network. However, even if the internal capacitor were as large as possible, the available space only allows for a 22 μ F 100 V ceramic capacitor in a 2220 package. Although this would decrease the V_{DS} overshoot compared to the existing 0.1 μ F version, there would still be a non-negligible resonance between the external and internal decoupling capacitors. Thus, the package inductances are still a source of unnecessary circulating currents causing additional losses and EMI noise.

Although these presented options would allow the exiting module to achieve an increased switching current capability, they diminish the potential performance that GaN power transistors offer by increasing complexity, size, cost, losses, and/or sacrificing reliability. To achieve high switching current capability, small switching losses, high switching frequency operation, reduced EMI noise generation, and high gate drive reliability, a module with improved parasitics is necessary.

III. GAN MODULE DESIGN ANALYSIS, EXAMPLES, AND RECCOMENDATIONS

In this section, GaN module layout recommendations are provided by analyzing the design of the commercial module as well as designs from modules presented in [11], [12], and [13].

A. Layout Analysis of a Commercial GaN Module

To analyze the design of the commercial GaN module, the casing was removed to reveal the layout, shown in Fig. 9. It was revealed that this module uses a dual bonded copper (DBC) substrate and wire bonds for die and trace connections. To better understand the electrical implications of this layout, the traces were modeled in 3D electromagnetic simulation software Computer Simulation Technology (CST). Fig. 10 shows the 3D model, developed using measurements of the module's traces. Using this model, three major aspects of the layout are analyzed, including: 1) lateral inner power loops, 2) cascaded layout of the parallel phases, and 3) use of wire bonds.

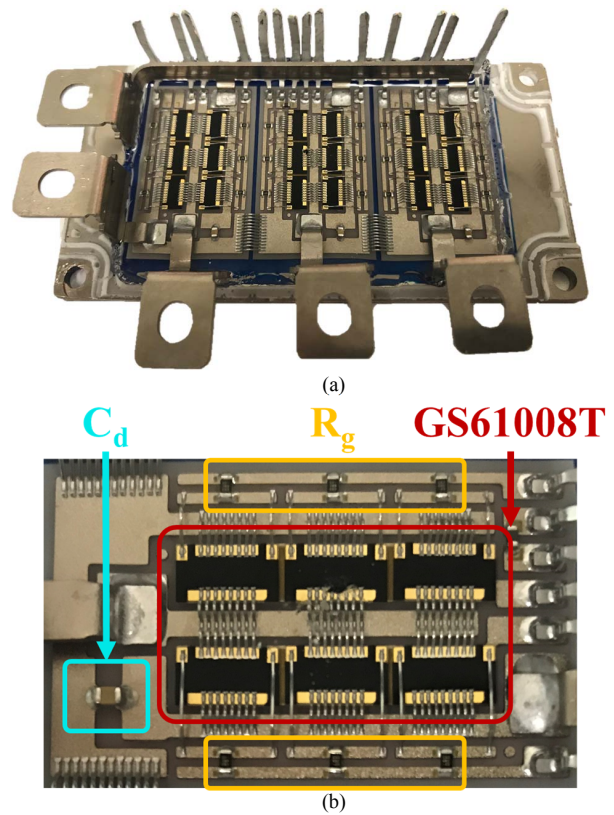


Fig. 9. (a) View of exposed commercial GaN module
(b) Close up view of a single phase

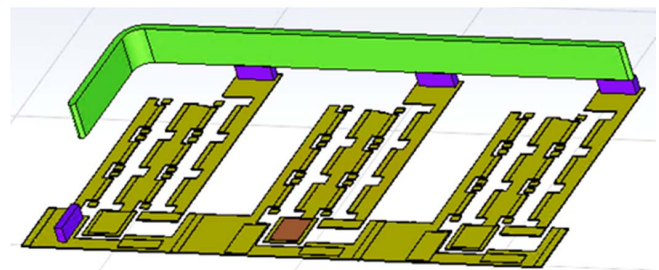


Fig. 10. CST model of the commercial GaN module's traces

1) Layout Analysis: Lateral Power Loops

The first aspect of the design that was analyzed is the use of a lateral inner power loop (loop including the internal local decoupling capacitor and its adjacent dies). This loop inductance is the major cause for the 70 MHz V_{DS} oscillation observed in the module hardware testing. To quantify the consequences of this layout, the loop inductance was simulated for each set of parallel dies (from closest to furthest from the decoupling capacitor). Fig. 11 shows the 3D model used for this simulation. For simplicity, the wire bonds and dies were modeled as continuous flat traces. The resulting loop inductances for the closest to furthest set of parallel dies were 11.4 nH, 19.8 nH, and 28.0 nH.

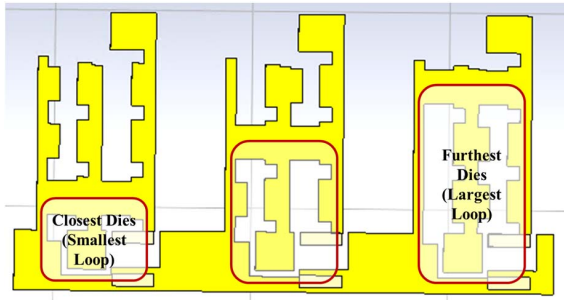


Fig. 11. CST model of the module's lateral power loop inductances

There are a couple points to make from this information. The first pertains to the asymmetry of the power loop inductances for the parallel dies. References [7] and [8] discuss the consequences of asymmetrical power loop inductances for paralleled GaN dies. The result is that the current sharing and switching characteristics aren't equal among all paralleled dies. This overstresses certain dies and underutilizes others, resulting in non-uniform V_{DS} overshoots, losses, and heat distribution.

Secondly, it has been shown that a lateral power loop (meaning the decoupling capacitor, dies, and connecting traces lie in the same plane) is not optimal for GaN applications due to its relatively large loop inductance. This is especially true for paralleled dies, as the further die in the lateral power loop will inherently have a larger and asymmetrical loop inductance. Instead, both EPC and GaN Systems recommend for their designs the use of flux canceling traces to achieve symmetric and minimized power loop trace inductances [9] [10]. Additionally, [9] discusses specifically how their recommended layout has significantly lower loop inductance than a lateral power loop. Although it is easiest and most effective to implement flux canceling traces using a PCB, it can be achieved using DBC or an Insulated Metal Substrate (IMS), which will be discussed in following sections.

2) Layout Analysis: Cascaded Layout of Parallel Phases

The second design aspect to be analyzed is the cascaded layout of the three parallel phases. Fig. 12 helps visualize the inductive loops created by this cascaded structure. These three inductive loops resonate with the internal $0.1 \mu\text{F}$ capacitors and cause the 2.1 MHz V_{DS} oscillation observed during hardware testing. Based on the parasitics model derived in Section II, the cumulative loop inductances from the module's +DC to -DC tabs through the first to third phase's internal decoupling capacitors are 29.1 nH , 51.8 nH , and 74.5 nH .

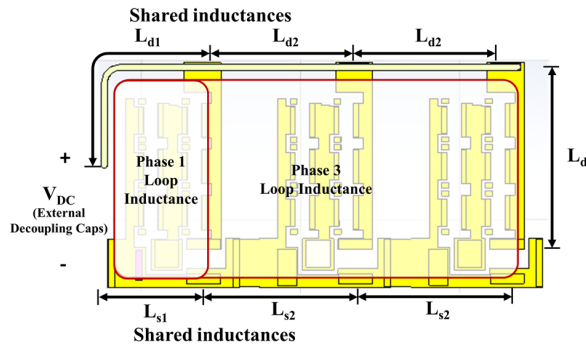


Fig. 12. Illustration of cascaded structure of parallel phases

There are two major issues related to this layout structure. Firstly, since the DC inputs to the module are not distributed bus bars, the loop inductances for each phase are inherently not equal. As discussed earlier, this creates non-uniform current sharing and resonances, overstressing certain dies and underutilizing the remaining. Secondly, since the cascaded phases have sections of shared parasitic inductances (namely L_{d1} , L_{s1} , and the leftmost L_{d2} and L_{s2} in Fig. 12), the switching operations of one phase directly impacts the other phases.

To illustrate this, the simulation model from Section II was implemented such that only the third phase was switching with an output DPT current of 90 A (one third of the third phase's current rating) and the first two phase's devices were kept off with their switch nodes connected to ground. Fig. 13 shows operation waveforms of this simulation. As shown, when only the third phase is switching there is still an induced V_{DS} spike and oscillations on the first two phase's devices. This causes additional losses and is a direct effect of the shared inductances of each phase's power loops.

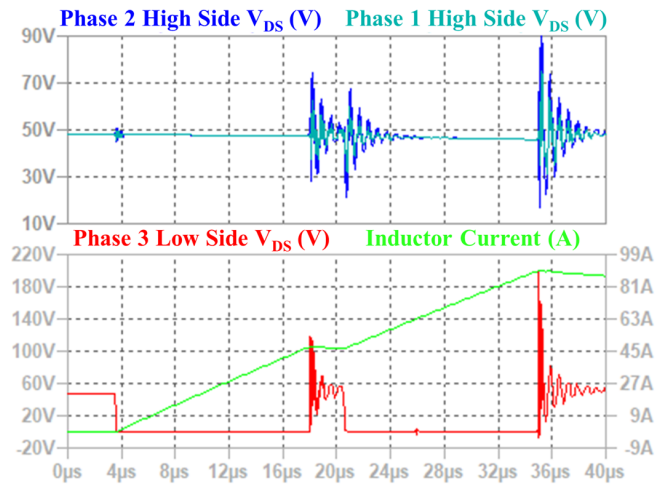


Fig. 13. Simulation results showing the effects of shared power loop inductances of parallel phases

The solution to both of these issues is to implement a DC bus input to the module with a distributed capacitance bank, such as the module shown in Fig. 14 [11]. A distributed bus like this would provide symmetrical power loop inductances for all parallel phases with no or minimized shared inductance between the parallel phases.

3) Layout Analysis: Wire Bonds

Wire bonds have been a common interconnect method in power modules for both power and signal traces. Unfortunately, wire bonds are one of the major sources of failure, increase inductance, and can potentially limit the module's current handling capability [11]. The inductance introduced by wire bonds can be on the scale of several nanohenries, which as demonstrated is non-negligible for GaN operation. Additionally, wire bonds inherently cannot be implemented in a flux canceling fashion. Thus, a wire bondless assembly has the potential to achieve smaller loop inductances than a wire bonded alternative, especially if trace flux cancelation is utilized.

B. Layout Recommendations from Existing GaN Modules

One of the first design choices when designing a module is the selection whether to implement the dies on DBC, IMS, or PCB (or alternative). The most common approach is to use DBC with a thick baseplate for thermal spreading and mechanical stability. Fig. 14 shows a GaN module (half bridge, 650 V, 150 A) designed using this approach from [11]. Their design paralleled five GS66508 dies with a claimed inductance between the +DC and -DC inputs of 5.1 nH to 5.6 nH (frequency dependent).

The low power loop inductance can be attributed to the low package profile, wide busbar and power traces, and the absence of wire bonds in the power loop. Additionally, the distributed DC busbar inputs help keep the power loop inductance for each set of paralleled dies symmetric. One limitation of this design relates to the gate drive design. As seen, the gate drive circuitry inputs are located at one end of the module. Presumably, this means that the gate loop inductance for each set of paralleled dies will be asymmetric and relatively large for the dies placed furthest from the gate drive circuitry. Also, it was stated that wire bonds were used for the die gate and source return connections, introducing additional gate loop inductance.

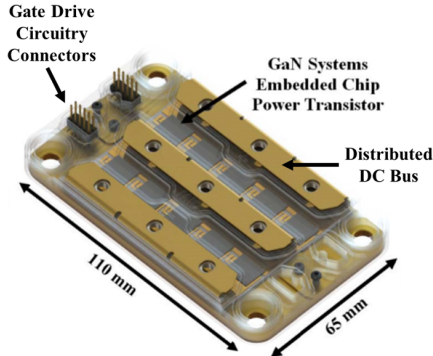


Fig. 14. DBC based GaN module design from [11]

Designed in a less traditional fashion, Fig. 15 shows a GaN module (half bridge, 650 V, 120 A) implemented on IMS from [12]. This module has four parallel GS66508B dies and integrates the power stage decoupling capacitors and gate driver ICs on a board elevated directly above it. The distributed power stage decoupling capacitors allow each paralleled set of dies to have symmetric power loop inductances for uniform current sharing and thermal performance. Additionally, the close proximity of the gate driver board allows for partial loop flux cancelation of the power and gate loop traces, resulting in a claimed commutation loop inductance of 1.3 nH, a quasi-common source inductance of < 0.3 nH, and gate driver loop inductances ranging from 6.8 nH to 7.4 nH (slight asymmetry).

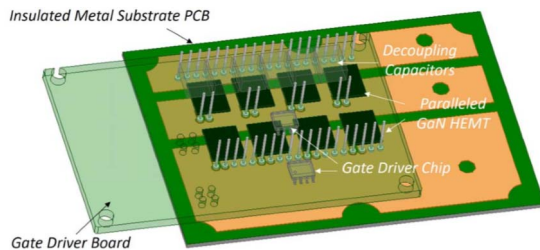


Fig. 15. IMS based GaN module design from [12]

The final module to be discussed is shown in Fig. 16 from [13]. This module (half bridge, 650 V, 240 A) is implemented on a 6-layer PCB and parallels four GS66516T dies. A high level of integration was achieved with the use of a PCB since it allows for the dies, power stage decoupling capacitors, and gate driver circuitry to all reside on a single board. Additionally, the close proximity of the PCB layers allows for optimal flux cancelation of the power loop traces, resulting in a claimed commutation loop inductance of 0.7 nH and quasi-common source inductance of < 0.2 nH. This low inductance design allowed for a DPT demonstration at its full current rating of 240 A while operating with a DC bus voltage of 400 V, achieving a V_{DS} overshoot of only 52 V.

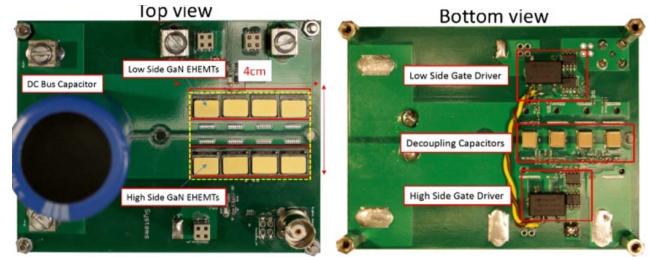


Fig. 16. PCB based GaN module design from [13]

IV. DEMONSTRATION OF A 100 V, 360 A GAN MODULE

To demonstrate the feasibility of a low voltage, high current GaN module, a prototype with four parallel GS61008 dies was developed, resulting in a 100 V, 360 A rated module. Fig. 17 shows a picture of the assembled DPT hardware setup. The dies were implemented on a 6-layer PCB in a low inductance flux canceling design [13]. Key hardware parameters were: $L = 2$ μ H, $V_{DC} = 48$ V, $R_{g(on)} = 10$ Ω (per die), $R_{g(off)} = 4$ Ω (per die), $V_{GS(on)} = 6$ V, and $V_{GS(off)} = -3.3$ V.

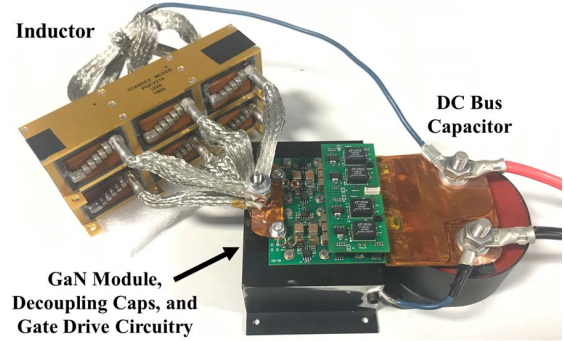


Fig. 17. Hardware DPT setup of a custom 100 V, 360 A GaN module

Fig. 18 shows the DPT waveform when hard switching up to 367 A. The first pulse's hard turn-off was at 200 A, and the V_{DS} peaked at 98.7 V and achieved a switching speed of 34.5 V/ns. The second pulse's hard turn-off was at 367 A, and the V_{DS} peaked at 85.5 V with a turn-off switching speed of 20.4 V/ns. This result was surprising, as the V_{DS} overshoot was expected to increase with an increasing current. Although it makes sense that the V_{DS} overshoot is less since the switching speed was slower, more work is needed to understand the mechanism causing the reduced switching speed of the second pulse. This will be reported in a future work.

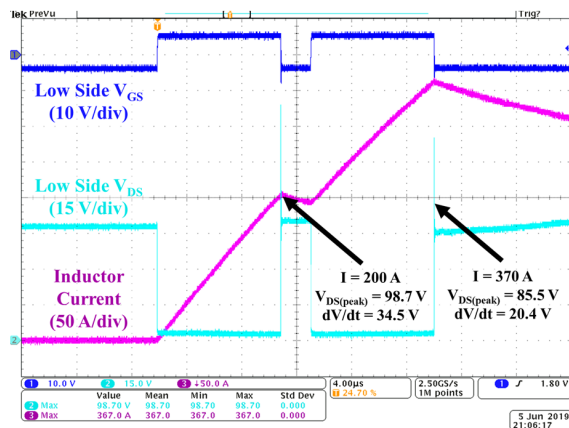


Fig. 18. Custom GaN module DPT waveforms at 367 A

In an attempt to further characterize the module, it was subjected to a 375 A single pulse (same hardware setup). The waveforms are shown in Fig. 19. As seen, the V_{DS} reached a peak of 109.5 V with a measured switching speed of 35.0 V/ns. Additionally, there is no low frequency oscillation on the V_{DS} waveform (such as the 2.1 MHz oscillation observed in the commercial module's DPT) due to the optimal power stage layout inspired by [13].

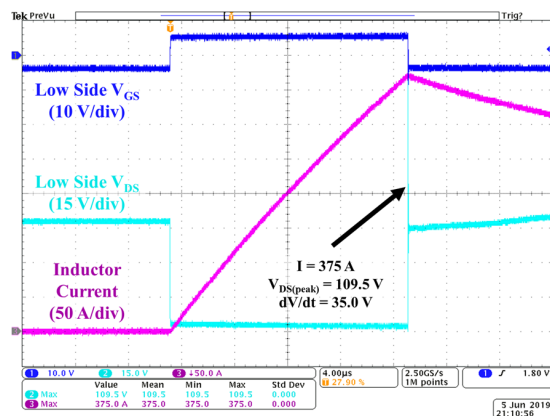


Fig. 19. Custom GaN module single pulse waveforms at 375 A

As demonstrated, this module was capable of hard switching at currents surpassing its die current ratings and with V_{DS} switching speeds reaching 35 V/ns. Compared to the commercial module analyzed in this paper, the custom module successfully switched at a 36% larger current (375 A compared to 277 A), had a >20% reduction in V_{DS} overshoot at that higher current (109.5 V compared to > 137.5 V), achieved faster switching (gate resistance of 4 Ω per die compared to 6.8 Ω per die), and used less than half of the parallel dies (4 parallel GS61008 dies compared to 9), successfully demonstrating the feasibility of a low voltage, high current GaN module and the benefits of minimized layout parasitics.

V. CONCLUSION

In this paper, the analysis of a commercial 3-phase, 100 V, 270 A (per phase) GaN module was presented. A detailed

parasitics model was derived and used to predict the premature failure of the module due to V_{DS} overshoot, a unique susceptibility for low voltage, high current GaN modules. This was confirmed with a hardware DPT setup, demonstrating module failure due to V_{DS} overshoot at 36% of its die current rating (failed at 292 A hard switching, rated for 810 A). Methods to work with the existing module were explored, demonstrating that optimal performance is only achievable with improved module parasitics. GaN module layout recommendations were presented based on analysis of the commercial module's layout as well as layouts from [11], [12], and [13]. The paper concluded with a hardware DPT demonstration of a custom 100 V, 360 A GaN module. The hardware was successfully hard switched at 375 A with a DC bus of 48 V, achieving a V_{DS} switching speed of 35 V/ns. This demonstrated the feasibility of a low voltage (< 100 V), high current (> 300 A) GaN module.

ACKNOWLEDGMENT

Mainstream Engineering Corp. would like to thank the US Army CCDC Ground Vehicle Systems Center for supporting this work under SBIR contract number W56HZV-18-C-0021.

REFERENCES

- [1] Q. Tao, J. Graham, J. Sun, "Characterization of IGBT modules for system EMI simulation", Proc. 25th Appl. Power Electron. Conf. Exp., pp. 2220-2225, 2010-Feb. 21-25
- [2] Lemmon, A., and Graves, R.: 'Parasitic extraction procedure for silicon carbide power modules'. Proc. Int. Workshop on Integrated Power Packaging, Chicago, USA, May 2015, pp. 91-94
- [3] Yang, L., and Odendaal, W.: 'Measurement-based method to characterize parasitic parameters of the integrated power electronics modules', Trans. Power Electron., 2007, 22, (1), pp. 54-62
- [4] GaN Systems, "GS61008T Top-side cooled 100 V E-mode GaN transistor," GS61008T datasheet, 2018.
- [5] GaN Systems, "GaN Switching Loss Simulation using LTSpice," 2018.
- [6] B. Hu, J. A. Brothers, X. Zhang, L. Fu, Y. M. Alsmadi and J. Wang, "An Isolated Phase-Shift-Controlled Quasi-Switched-Capacitor DC/DC Converter With Gallium Nitride Devices," in IEEE Journal of Emerging and Selected Topics in Power Electronics, vol. 7, no. 2, pp. 609-621, June 2019.
- [7] EPC, Reusch, D., "Effectively Paralleling Gallium Nitride Transistors for High Current and High Frequency Applications," 2016.
- [8] N. Sang, P. Jeannin and P. Lefranc, "Analyses of the unbalanced paralleled GaN HEMT transistors," 2018 20th European Conference on Power Electronics and Applications (EPE'18 ECCE Europe), Riga, 2018, pp. P.1-P.9.
- [9] EPC, Reusch, D., "Optimizing PCB Layout," 2014.
- [10] GaN Systems, "Design considerations of Paralleled GaN HEMT-based Half Bridge Power Stage," 2018.
- [11] B. Passmore et al., "A 650 V/150 A enhancement mode GaN-based half-bridge power module for high frequency power conversion systems," 2015 IEEE Energy Conversion Congress and Exposition (ECCE), Montreal, QC, 2015, pp. 4520-4524.
- [12] J. L. Lu, D. Chen and L. Yushyna, "A high power-density and high efficiency insulated metal substrate based GaN HEMT power module," 2017 IEEE Energy Conversion Congress and Exposition (ECCE), Cincinnati, OH, 2017, pp. 3654-3658.
- [13] J. L. Lu and D. Chen, "Paralleling GaN E-HEMTs in 10kW-100kW systems," 2017 IEEE Applied Power Electronics Conference and Exposition (APEC), Tampa, FL, 2017, pp. 3049-3056.